

Binomial expansion - Set 1 - Solutions

Specification MM2.4.

Answer key

Question	Answer	Topic
1	B	Binomial expansion
2	A	Binomial expansion
3	D	Binomial expansion
4	A	Binomial expansion
5	D	Binomial expansion
6	D	Binomial expansion
7	B	Binomial expansion
8	B	Binomial expansion
9	E	Binomial expansion
10	D	Binomial expansion
11	H	Binomial expansion
12	A	Binomial expansion

Worked solutions

Question 1 BIN-01 • Binomial expansion • spec MM2.4

Find the coefficient of x^3 in the expansion of $(2 + 3x)^5$.

Answer: B

$$\binom{5}{3}2^2(3x)^3 = 10 \times 4 \times 27x^3 = 1080x^3.$$

Question 2 BIN-02 • Binomial expansion • spec MM2.4

The coefficient of x^2 in the expansion of $(1 + kx)^6$ is 60. Given that $k > 0$, find k .

Answer: A

$$\binom{6}{2}k^2 = 15k^2 = 60, \text{ so } k^2 = 4 \text{ and } k = 2.$$

Question 3 BIN-04 • Binomial expansion • spec MM2.4

What is the sum of the coefficients in the expansion of $(3x - 2)^7$?

Answer: D

$$\text{Put } x = 1: (3 - 2)^7 = 1.$$

Question 4 BIN-05 • Binomial expansion • spec MM2.4

Evaluate $\binom{10}{3} - \binom{10}{7} + \binom{9}{2}$.

Answer: A

$$\binom{10}{3} = \binom{10}{7}, \text{ so the first two cancel; } \binom{9}{2} = 36.$$

Question 5 BIN-06 • Binomial expansion • spec MM2.4

In the expansion of $(1 + x)^n$, the coefficients of x^4 and x^5 are equal. Find n .

Answer: D

$$\binom{n}{4} = \binom{n}{5} \text{ requires } 4 + 5 = n \text{ (by symmetry } \binom{n}{r} = \binom{n}{n-r} \text{)}, \text{ so } n = 9.$$

Question 6 BIN-07 • Binomial expansion • spec MM2.4

Expand $(1 - 2x)^4$ and hence find the coefficient of x^2 in $(1 + 3x)(1 - 2x)^4$.

Answer: D

$$(1 - 2x)^4 = 1 - 8x + 24x^2 - 32x^3 + 16x^4. \text{ Coefficient of } x^2 \text{ in the product: } 1 \cdot 24 + 3 \cdot (-8) = 0.$$

Question 7 BIN-10 • Binomial expansion • spec MM2.4

Find the coefficient of x^5 in the expansion of $(1 + x)^3(1 + x)^4$.

Answer: B

$$(1 + x)^3(1 + x)^4 = (1 + x)^7, \text{ and } \binom{7}{5} = 21.$$

Question 8 BIN-13 • Binomial expansion • spec MM2.4

Use the binomial expansion to find the value of 1.01^5 correct to 4 decimal places.

Answer: B

$$(1 + 0.01)^5 = 1 + 5(0.01) + 10(0.0001) + 10(0.000001) + \dots = 1 + 0.05 + 0.001 + 0.00001 + \dots \approx 1.0510.$$

Question 9 BIN-03 • Binomial expansion • spec MM2.4

Find the term independent of x in the expansion of $\left(x^2 + \frac{2}{x}\right)^6$.

Answer: E

$$\text{General term } \binom{6}{r}(x^2)^{6-r}\left(\frac{2}{x}\right)^r = \binom{6}{r}2^r x^{12-3r}. \text{ Need } 12 - 3r = 0, r = 4: \binom{6}{4} \times 16 = 15 \times 16 = 240.$$

Question 10 BIN-08 • Binomial expansion • spec MM2.4

The first three terms in the expansion of $(1 + ax)^n$ in ascending powers of x are $1 + 12x + 54x^2$. Find n .

Answer: D

$$na = 12 \text{ and } \frac{n(n-1)}{2}a^2 = 54. \text{ Dividing the second by the square of the first: } \frac{n-1}{2n} = \frac{54}{144} = \frac{3}{8}, \text{ so } 8n - 8 = 6n \text{ and } n = 4 \\ \text{(then } a = 3; \text{ check } \binom{4}{2} \cdot 9 = 54).$$

Question 11 BIN-09 • Binomial expansion • spec MM2.4

Which of the following statements is/are true?

- I** $\binom{n}{r} = \binom{n}{n-r}$ for all integers $0 \leq r \leq n$
- II** $\binom{n}{r} + \binom{n}{r+1} = \binom{n+1}{r+1}$ for all integers $0 \leq r < n$
- III** $\sum_{r=0}^n \binom{n}{r} = 2^n$ for all positive integers n

Answer: H

I is the symmetry of Pascal's triangle; II is Pascal's rule (each entry is the sum of the two above it); III is $(1 + 1)^n$ expanded. All true.

Question 12 BIN-11 • Binomial expansion • spec MM2.4

Find the value of $(\sqrt{2} + 1)^4 + (\sqrt{2} - 1)^4$.

Answer: A

$$\text{Odd powers of } \sqrt{2} \text{ cancel: } 2[(\sqrt{2})^4 + \binom{4}{2}(\sqrt{2})^2 + 1] = 2[4 + 12 + 1] = 34.$$